

BROWNSVILLE, TX, USA

WMO: 722500

Lat: 25.9167N

Lon: 97.4192W

Elev: 8

StdP: 101.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.7	5.8	-4.0	2.7	10.5	-1.1	3.4	10.5	13.1	25.2	12.0	24.9	4.9	330	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	35.6	25.8	34.9	25.7	34.3	25.7	27.5	31.3	27.1	31.0	26.8	30.8	6.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.3	28.7	26.2	21.7	28.5	25.8	21.1	28.3	87.2	31.4	85.3	31.0	83.8	30.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.8	9.4	DB	1.0	37.6	2.0	1.3	-0.4	38.6	-1.6	39.3	-2.7	40.0	-4.2	41.0	
			WB	-0.6	28.2	1.9	0.7	-1.9	28.7	-3.0	29.1	-4.1	29.4	-5.5	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.9	16.6	18.6	21.0	24.1	27.1	29.1	29.5	29.8	27.9	24.9	20.9	17.4
	DBStd	5.82	4.93	4.86	4.00	3.32	2.29	1.37	1.19	1.29	1.95	3.31	4.33	4.93
	HDD10.0	21	9	4	1	0	0	0	0	0	0	0	1	6
	HDD18.3	277	91	53	22	4	0	0	0	0	0	3	25	79
	CDD10.0	5108	214	244	341	423	532	574	604	614	538	461	329	234
	CDD18.3	2323	37	59	104	177	273	324	346	356	288	206	103	49
	CDH23.3	23456	166	289	553	1332	2749	3874	4279	4390	3022	1844	703	253
Wind	WSAvg	4.6	4.5	5.0	5.2	5.6	5.6	4.9	4.9	4.2	3.3	3.8	4.0	4.2
	PrecAvg	671	33	27	23	37	66	67	42	64	147	85	39	32
	PrecMax	1193	122	120	151	168	247	226	240	243	513	348	143	143
	PrecMin	400	1	0	0	0	0	0	0	0	13	0	0	0
	PrecStd	168	29	27	31	40	56	57	46	51	105	75	34	28
	DB	28.1	30.0	31.3	33.6	34.7	35.9	36.2	36.6	35.9	33.6	31.1	28.9	
	MCWB	20.7	20.8	22.6	23.2	25.6	26.1	25.8	26.1	25.3	25.5	23.7	22.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	26.7	28.0	29.1	31.7	33.4	34.7	35.2	35.7	34.3	32.4	29.6	27.4	
	2% MCWB	20.6	21.9	22.2	23.3	25.2	25.9	25.7	25.9	25.4	25.1	23.1	22.1	
	5% DB	25.3	26.6	27.9	30.4	32.4	34.0	34.5	35.0	33.3	31.3	28.5	26.2	
	5% MCWB	19.8	21.2	21.7	23.4	24.9	25.7	25.7	25.8	25.5	24.6	22.6	21.4	
	10% DB	23.7	25.1	26.5	29.1	31.4	33.2	33.8	34.3	32.3	30.2	27.3	24.6	
	10% MCWB	19.2	20.2	21.0	22.8	24.7	25.6	25.7	25.8	25.4	24.0	22.0	20.7	
	WB	22.8	23.5	24.2	25.9	27.3	28.2	27.6	28.0	27.7	27.4	25.7	24.0	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	25.7	26.9	28.4	30.1	31.1	32.9	31.7	31.9	30.8	30.8	28.4	26.7	
	2% WB	21.9	22.8	23.5	25.1	26.7	27.5	27.0	27.4	27.0	26.7	24.8	23.2	
	2% MCDB	24.5	25.8	27.1	29.2	30.6	30.9	31.2	31.7	30.5	29.9	27.4	25.6	
	5% WB	21.2	22.2	22.9	24.5	26.1	26.9	26.7	26.9	26.7	26.0	24.1	22.5	
	5% MCDB	23.8	24.9	26.0	28.3	29.9	30.6	31.0	31.3	30.4	29.1	26.7	24.9	
	10% WB	20.5	21.6	22.3	24.0	25.6	26.6	26.4	26.6	26.3	25.4	23.3	21.7	
	10% MCDB	22.9	24.0	25.0	27.4	29.2	30.4	30.7	31.1	30.1	28.6	26.0	24.0	
Mean Daily Temperature Range	MDBR	10.1	9.8	9.3	9.1	8.4	8.6	8.8	9.3	8.8	9.5	9.9	10.0	
	5% DB	10.7	9.9	9.6	9.7	9.0	9.3	9.7	10.0	9.7	9.7	9.9	10.1	
	MCWBR	5.6	4.4	3.9	3.6	2.5	2.4	2.2	2.2	2.9	3.7	4.5	5.6	
	5% WB	9.1	8.6	8.3	8.5	7.5	8.0	8.6	9.0	8.1	8.0	8.4	8.7	
	MCWBR	5.5	4.4	3.8	3.5	2.4	2.4	2.4	2.4	3.0	3.7	4.7	5.5	
Clear-Sky Solar Irradiance	taub	0.368	0.381	0.402	0.450	0.463	0.434	0.453	0.439	0.441	0.393	0.379	0.371	
	taud	2.464	2.418	2.355	2.223	2.218	2.364	2.320	2.367	2.330	2.461	2.458	2.456	
	Ebn at Noon	883	897	894	855	839	857	841	854	845	873	863	862	
	Edh at Noon	98	110	123	144	145	124	130	123	125	104	98	96	
All-Sky Solar Radiation	RadAvg	3.12	3.77	4.64	5.52	6.14	6.62	6.57	6.31	5.25	4.60	3.53	2.83	
	RadStd	0.34	0.53	0.47	0.44	0.35	0.23	0.44	0.25	0.35	0.40	0.36	0.38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+148	
Regional (12 neighbors)	N/A	N/A	N/A	+0.31	N/A	N/A	N/A	N/A	N/A	+126	

Nomenclature: See separate page